Two roads, one tuning.

Start: **išartum**, after 7 earlier moves. Destination: **embūbum**.

Initial session string 1: 440 Hz. Companion strings: octave. Sound: plucked.

Modern exact 4/3 and 3/2 realization. These numerical pitches and the comparison transposition are not recorded on the tablets.

A · Loosen (2 moves)

1. Loosen 2 + 9 **kitmum**
2. Loosen 6 **embūbum**

B · Tighten (5 moves)

1. Tighten 5 **qablītum**
2. Tighten 1 + 8 **nīš tuḫrim**
3. Tighten 4 **nīd(i) qablim**
4. Tighten 7 **pītum**
5. Tighten 3 **embūbum**

Same name, different pitches

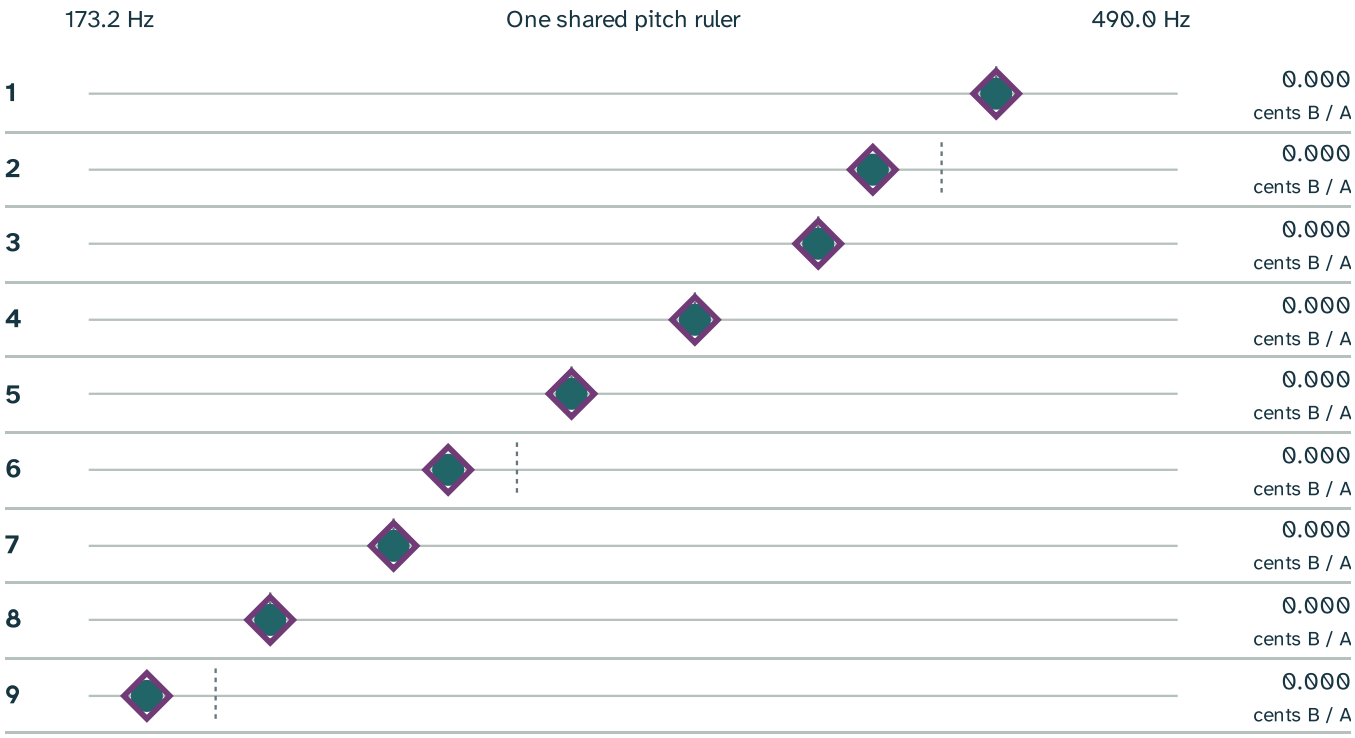
Every raw B pitch is 2187/2048 times A, a shift of 113.685006 cents. The interval pattern matches exactly on all nine strings.

Listen to the comparison

Route A, then route B: nine strings; B transposed to match A on string 1.

Only the heard B is transposed, by 2048/2187. The raw B column keeps the retuned result.

Filled circle: A. Open diamond: heard B. Dashed tick: starting pitch. Every string shares one logarithmic ruler.



Check every string

Fractions refer to the original session's initial string 1. The two raw results stay separate from heard B.

Retuned pitches and the chosen listening comparison

String	A: ratio / Hz	B: ratio / Hz	B / A	Heard B
1	2048/2187 412.034751 Hz	1/1 440.000000 Hz	2187/2048	412.034751 Hz
2	16384/19683 366.253112 Hz	8/9 391.111111 Hz	2187/2048	366.253112 Hz
3	64/81 347.654321 Hz	27/32 371.250000 Hz	2187/2048	347.654321 Hz
4	512/729 309.026063 Hz	3/4 330.000000 Hz	2187/2048	309.026063 Hz
5	4096/6561 274.689834 Hz	2/3 293.333333 Hz	2187/2048	274.689834 Hz
6	32768/59049 244.168741 Hz	16/27 260.740741 Hz	2187/2048	244.168741 Hz
7	128/243 231.769547 Hz	9/16 247.500000 Hz	2187/2048	231.769547 Hz
8	1024/2187 206.017375 Hz	1/2 220.000000 Hz	2187/2048	206.017375 Hz
9	8192/19683 183.126556 Hz	4/9 195.555556 Hz	2187/2048	183.126556 Hz

Keep the experiment

Reopen this exact comparison

Text, reconstruction, calculation

The seven-name network follows the scholarly reconstruction. Mirelman (2013), figure 5, p. 52 and pp. 53–55 discusses the pattern and interpretation. Rahn (2022), paragraphs 11–16 treats the reversible network and repeated cycles; paragraph 45 explains the editorial basis of tightening terminology. The two routes' exact frequency factor and their transposed equality are deductions from this workshop's chosen 4/3 and 3/2 realization, not recovered ancient measurements.

nê'um: loosen. nasāḫum: the contextual tightening reading, in an extensively restored chapter. These are normalized lemmas, not a reconstructed ancient sentence or pronunciation.

- Sam Mirelman · Tuning Procedures in Ancient Iraq (2013)
- Jay Rahn · Was Mesopotamian Tuning Diatonic? A Parsimonious Answer (2022)

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